Koalified system architecture

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Overview

Koalified v1 is an iPhone app that keeps everything on the student's phone: they bring their own timetable and marks, and nothing personal is sent to a server. This is the safest way to launch, and it matches UQ's rules on credentials and student data.

Area	Include in v1	Leave out for now
Data	Stored on the phone only	Any personal data on a server
Login	None	UQ credentials (never); app accounts (later)
Sources	Timetable file (.ics), Excel sheet, hand-entered degree plans, public facts	Any scraping, copying course content
Features	Timetable, deadlines, marks, what-if calculator, degree plan, semester calendar, widget, goals, help	Plus, Momentum, Shared Fires, profile sharing, study groups
Backend	None	Accounts, sync, push server, verified-deadline service

Scope

Build the ten must-have sections first. Everything in the upcoming group waits until the first 100 users are in. The names match the sections in the Figma file.

Feature	Version	Needs a server?	Figma section
Onboarding, import, app tour	v1	No	MVP 01
Today, next class, class details	v1	No	MVP 02
Courses, marks, hurdle	v1	No	MVP 03
Degree plan	v1	No (bundled data)	MVP 04
What-if calculator	v1	No	MVP 05

Feature	Version	Needs a server?	Figma section
Goals	v1	No	MVP 06
Excel import and import check	v1	Only to receive reports	MVP 07
Semester calendar	v1	No (bundled dates)	MVP 08
Profile, settings, Home Screen widget	v1	No	MVP 09
Help, FAQ, feedback, roadmap	v1	Only to receive reports	MVP 10
Study coach, planner, GPA forecast (Plus)	Later	Yes	Upcoming 11
Momentum and Shared Fires	Later	Yes (buddies)	Upcoming 12
Profile sharing, verified deadlines	Later	Yes	Upcoming 13

Do not build:

- Anything that asks for a UQ username or password.
- Any scraping in v1, public pages included. Never scrape behind a login (myUQ, Blackboard, my.UQ timetable).
- A database of other students' grades, enrolments or timetables.
- Copies of course profile text or Library content.

UI and client app

The app is four tabs with a floating tab bar: Today, Courses, Calculator and Profile. The Figma file already has 40 or so screens and a clickable prototype covering these.

- **Navigation:** floating tab bar; sheets for adding marks, goals and reports; a first-run tour of 4 steps.
- **Design system:** Inter (Regular and Semi Bold only), 13 text styles, purple, white and black, rounded cards with soft shadows. Green and red are used only for the exam hurdle result.
- **Widgets:** small next-class widget and a medium Today widget, standard (free) in v1.
- **Notifications:** local reminders only (deadlines, classes). No push server in v1.
- **Accessibility:** minimum text size 11 px, layouts must grow with the device text size setting, and no meaning carried by colour alone.

- **Placeholder content:** every course, date and program in Figma is sample data and must be replaced by real data before launch.

Decision	Recommendation	Why
Platform	iPhone first	One platform, best widgets, App Store target
Dark mode	Not in v1	Doubles design and test work
Languages	English only	Small team of one

Interface rules

These rules come from the design review and apply to every screen in the Figma file.

- **Colour:** neutral grey canvas, purple only for primary buttons, links and the selected state.
- **Text levels:** the task or event name is 14 semi-bold. Codes, weights and rooms sit underneath in 11 regular grey.
- **Navigation:** four tabs (Today, Courses, Calculator, Profile). A quiet floating bar with no outline. Setup and import flows hide it and show Back, Cancel and step progress.
- **Today:** next-class card first, then an agenda grouped as Today, Tomorrow and This week. Deadlines come before goals and streaks. An Agenda and Calendar switch replaces the calendar button.
- **Courses:** a switch between My courses and Degree plan. Semester filters stay inside My courses.
- **Empty states:** "nothing scheduled" and "no timetable yet" are different screens.
- **Calculator:** target-grade pills give the needed mark in plain English with the assumption in the same sentence. One slider and a typed box fine-tune it. A failed hurdle replaces the grade with a warning. States exist for already secured, not reachable and weights that do not add up to 100%.
- **Accessibility:** touch targets at least 44 by 44 pt, auto layout so text can grow, a word or icon with every colour, dark-mode secondary text no darker than #B0B0B5.
- **Sheets and haptics:** editing a mark or adding a deadline opens a bottom sheet. A light haptic fires when the slider crosses a grade boundary. A success haptic fires after an import or a completed assessment.
- **Trust labels:** Official, Verified, Unverified and Sample each open a sheet showing the source and last-checked date. Adding a deadline asks whether it is just for me or shared with the course.

- **Onboarding:** there is no sign-up or login in version 1. Students pick their university, then import a timetable or set up by hand, with a sample Today available first. Accounts and sync are an upcoming feature. The multi-screen tour is replaced by two first-run tips.
- **Import problems:** a partly imported file shows what was read, what was left out and how to fix it.

Animation and motion

Focus animations on moments where data changes dynamically, state elevates, or the user experiences an emotional milestone. In native iOS, gratuitous screen-wide transitions feel laggy, whereas precise micro-animations make the app feel responsive and fluid.

1. The mascot reactivity (header transition)

- **Screen:** Today view and active sprint.
- **Transition:** switching between the passive idle state and the active study state (putting on headphones, resting chin on hand, looking down at notes).
- Use a subtle morph or cross-fade with `.spring(response: 0.45, dampingFraction: 0.75)` on the container shape.
- Animate the vertical translation (`.offset(y: ...)`) subtly rather than jumping abruptly when the user taps Start Sprint or Pause.

2. The what-if grade calculator and slider scrubbing

- **Screen:** Calculator and the final-exam target sheet.
- **Interaction:** moving the hypothetical grade slider back and forth.
- **Dynamic numeric rollover:** the projected percentage text (for example 78.6% to 81.2%) uses `.contentTransition(.numericText())` so the individual digits roll vertically like an odometer instead of flashing.
- **Grade threshold capsule pop:** when the percentage crosses into a new grade boundary (for example from Grade 5 to Grade 6), trigger a light spring scale animation.
- Pair this visual pop with a `.sensoryFeedback(.selection)` haptic tick.

3. Assessment check-off and task dismissal

- **Screen:** Today's small wins and the assessment list.
- **Interaction:** tapping the circular radio or checkbox to complete an assessment or study milestone.
- **Checkmark pop:** scale from 0.0 to 1.2 before settling at 1.0.

- **Row strike-through and dim:** text shifts to `.secondary` opacity with a smooth ease-out.
- **Card collapse:** if completed items auto-dismiss to reveal "You're done for today" or "Everything submitted", animate the row collapsing with `.transition(.asymmetric(insertion: .identity, removal: .move(edge: .leading).combined(with: .opacity)))`.

4. Bottom sheets (adding marks and manual inputs)

- **Screen:** entering assignment marks or setting up hurdle rules.
- **Transition:** opening the edit dialog.
- Avoid full-screen navigation pushes (`NavigationLink`) for quick edits.
- Use a native interactive sheet: `.presentationDetents([.fraction(0.4), .medium])`.
- Enable `.presentationDragIndicator(.visible)` so the user can swipe it away with rubber-band physics.

5. The milestone and semester wrapped card carousel

- **Screen:** semester recap and the "Your Results, Your Progress" three-slide story.
- **Transition:** swiping between slide 1 (Focus), slide 2 (Deadlines) and slide 3 (Final Grades).
- Use a paging carousel: `TabView` with `.tabViewStyle(.page(indexDisplayMode: .never))`.
- Add a slight parallax or scale effect to the progress bar and summary cards so the numbers count up dynamically from 0 to 36 hours and 7 of 9 submitted as each card snaps into the center viewport.

6. Timeline card expansion (class countdown)

- **Screen:** Today dashboard schedule items (for example CSSE1001: 25 min left).
- **Transition:** expanding from a collapsed one-line event pill to the full card showing room location, building code and map details.
- Apply `matchedGeometryEffect` across the title, room badge and card background. This gives a fluid expansion where the card grows directly from its tapped position rather than popping open instantly.

Data and storage

All personal data lives in a local database on the phone. The only data that ships with the app is public, non-personal reference data.

Data	Where it lives	Leaves the phone?
Courses, classes, deadlines, marks, goals	Local database (SwiftData or SQLite)	No
Notification settings, preferences	Local settings	No
Degree plans, semester dates, course list	Bundled with the app, updated by app release	No (read only)
Imported files (.ics, Excel)	Read once, then discarded	No
Import reports and bug reports	Sent only when the student taps Send	Yes, by choice
Crash and usage data	Anonymous, no names or marks	Yes, anonymous

- **Backup:** the student's own iCloud device backup covers the local database. No Koalified backup in v1.
- **Delete:** a "Delete all my data" button in Profile wipes the local database.
- **Widgets:** read from the same local data through an App Group, so nothing is copied elsewhere.
- **Later (needs accounts):** sync, Shared Fires and verified deadlines add a small cloud database. Keep deadline facts separate from anything identifying a person.

Security and privacy

The strongest protection is not collecting the data. In v1 there is no server to breach and no credentials to leak.

Risk	Control in v1	Status
UQ credentials	Never requested, never stored; no login form	Required
Personal data exposure	On-device only; no analytics tied to a person	Required
Lost or stolen phone	Rely on iOS passcode and device encryption; optional Face ID lock	Optional
Imported files	Parsed locally, then deleted; never uploaded	Required
Reports sent by students	Show exactly what is sent; attaching a file is opt-in	Required

Risk	Control in v1	Status
Third-party code	Few dependencies; no ad or tracking SDKs	Required
Permissions	Notifications only (and Calendar if the student chooses to export)	Required

Apple requirements

- A privacy policy URL and App Privacy answers before submission.
- If accounts are added later: account deletion inside the app, and Sign in with Apple when other social logins are offered.

Later, with accounts and student-added deadlines

- Student-added deadlines need abuse limits: a confirmation threshold (3), rate limits per user, and official dates always override student ones.
- Never store names, student numbers or grades next to deadline data.

Data sources and ingestion

Version 1 uses no scraping of any kind. Data comes from files the student supplies or from data you enter by hand. Consider automated reading of public pages only after v1, once you have checked UQ's terms and have written permission.

Source	Method	Risk	Use in v1
Student's timetable	Import a .ics file they export themselves	Low	Yes
Student's marks	Excel import or manual entry	Low	Yes
Degree plans	Hand-entered from public program pages for the first programs	Low	Yes
Semester dates, holidays	Hand-entered from the public academic calendar	Low	Yes
Course assessment dates and hurdles	Student pastes text from their own course profile (read on the phone) or types it in	Medium	Student-entered
Public Programs and Courses pages	Automated reading	Medium: load limits, copyright	Not yet

Source	Method	Risk	Use in v1
myUQ, Blackboard, my timetable	Anything behind a login	Not allowed	Never
Library resources	Crawlers	Not allowed	Never

Rules if you later read public pages

- Ask UQ for written permission first.
- Keep only facts (dates, weights, hurdles), never the text of course profiles.
- Rate-limit hard, cache results, and refresh a course at most once a day.
- Link to the official page instead of copying it.

Import check loop: after every import the app asks "Does this look right?". A wrong answer logs which part is wrong (courses, class times, deadlines, degree plan, semester dates) so you can fix it for everyone.

Assessments and hurdles (the hurdle feature depends on this): with no scraping, each student sets up their courses. The "Set up this course" screens let them paste the assessment table from their own course profile, check the rows the app suggests, and switch on a hurdle with its minimum mark. They can also add rows by hand. Hurdles stay optional, so the calculator works without one. Starter data for your own courses can ship with the app; a shared, student-verified library comes later with accounts.

Course profile link or code search. A student can type a course code or paste a public course profile link, and the app reads that one page on their own phone when they ask. It stays behind a feature flag until UQ gives written permission. Pasting the profile text, or typing assessments by hand, stays available as the fallback.

Backend and services

Version 1 needs no backend of its own. Only the report forms need somewhere to send messages, and that can be an email address or a form tool.

Service	v1	Later	Notes
Bug and feedback intake	Email or a hosted form	A small API	Free tier is enough
Accounts and sign-in	None	Sign in with Apple	Needed for sync and buddies
Sync across devices	None	Cloud database	Adds privacy duties

Service	v1	Later	Notes
Verified deadlines	None	Small API + database	Course facts only
Push notifications	Local only	Push service	Needed for buddy nudges
Subscriptions (Plus)	None	StoreKit with a receipt service	Apple takes 15 to 30%
AI study coach	None	Hosted model API	Cost per user; price before launch

- **Rule of thumb:** add a server only when a feature cannot work without one.
- **Cost:** v1 should cost the Apple Developer fee (about US$99 a year) and nothing else.

Tools and tech stack

Choose the stack you can ship in a few Friday afternoons per week. The native iPhone route is the best fit for widgets and a clean App Store launch.

Layer	Recommendation	Alternative	Why
App framework	SwiftUI (Swift)	React Native or Flutter	Best widgets and iOS feel; cross-platform only if you want Android early
Local storage	SwiftData	SQLite with GRDB	Built in, no server
Widgets	WidgetKit with an App Group	None	The only supported route on iOS
Reminders	Local notifications	None	No server needed
File import	Apple's file importer, ICS parser, Excel reader	None	Everything stays on the phone
Design	Figma file (already built)	None	Source of truth for screens and styles
Source control	Git on GitHub	None	Backups and history
Builds and beta	Xcode Cloud or GitHub Actions + TestFlight	None	Free beta testing

Layer	Recommendation	Alternative	Why
Crash reporting	Apple's built-in crash reports	A third-party SDK	Avoids extra privacy duties
Analytics	Apple's App Analytics only	Privacy-friendly analytics	Weekly active users without tracking people

- **Skills fit:** SwiftUI is also a strong resume line for an iOS role.
- **Avoid in v1:** ad SDKs, social login SDKs, and any library that sends data off the phone.

Testing, release and App Store

Aim to submit in early February so the app is live in the first weeks of Semester 1, when students build their timetables.

1. **Unit tests** for the calculator maths, hurdle rule and grade bands, plus import parsing.
2. **Beta on TestFlight** with about 20 friends across different courses, then fix what the import check reports.
3. **Submit** with screenshots, description, support page and privacy policy.
4. **Launch** and measure weekly active users, not only downloads.

Before you submit	Done?
Apple Developer account (about US$99 a year)	Not yet
Privacy policy and support page published	Not yet
App name final and checked for clashes	Not yet
No UQ name or logo in the app or listing	Not yet
All sample data replaced with real data	Not yet
Artwork rights confirmed in writing	Not yet
Account deletion in the app (only if accounts exist)	Not needed in v1

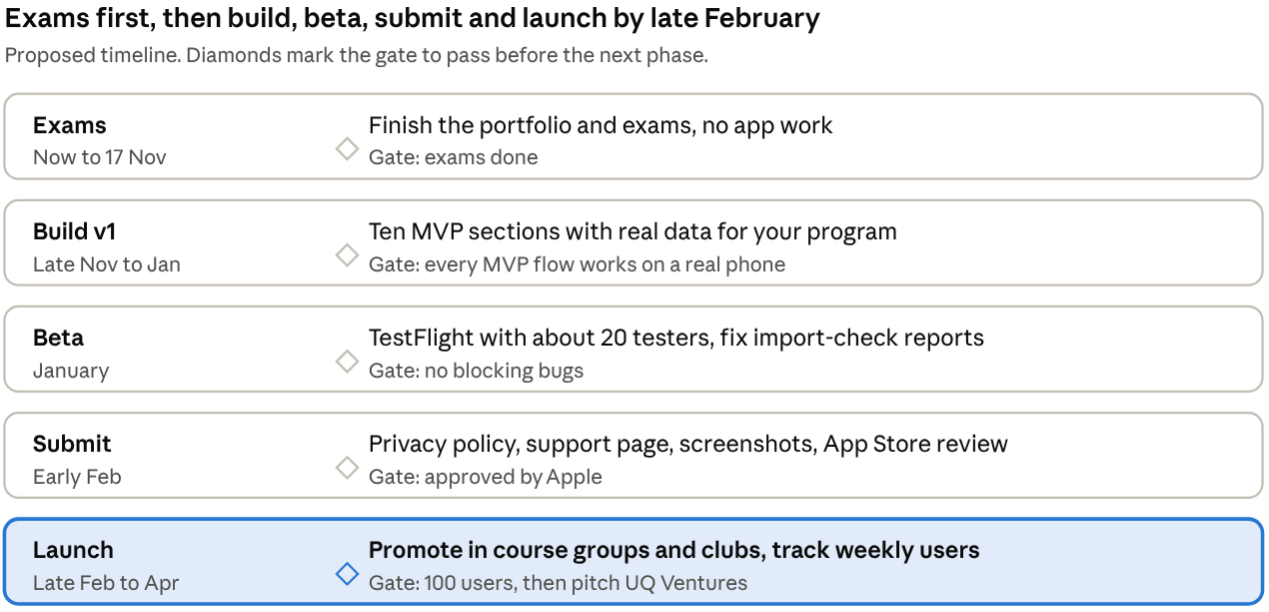
Legal and UQ compliance

The summary of UQ's rules you pasted was not checked against the original policies, so read the policies themselves before you rely on any clause number. The direction is clear: no credentials, no scraping behind a login, no holding other students' records.

Item	What to do	When
UQ ICT and data-handling policies	Read the originals in UQ's policy library	Before building anything automated
Permission to read public pages	Email Student Services or ITS and ask in writing	Before any scraping
UQ name and logo	Not used in the app or App Store listing	Always
Course content	Keep facts only; link to the official page	Always
Library resources	No crawling	Always
IP on a student project	Check UQ's policy; confirm the app is yours	Before the pitch
Friend's koala artwork	Get the licence or assignment in writing	Before launch
Privacy policy	Plain language, matches what the app really does	Before submission
Australian Privacy law	If you hold personal data later, follow the Privacy Act and Queensland rules	Before accounts or sync

This is planning information, not legal advice. For anything binding (IP, privacy, permission), ask UQ Ventures or a free student legal service.

Build order and roadmap



roadmap · 5 phases, 5 gates

Each phase must pass its gate before the next starts. Upcoming features (Plus, Momentum, sharing, verified deadlines) only start after the 100-user gate, because each needs accounts and a server.

Risks, open decisions and checklist

The biggest risk is time: exams run until 17 Nov and you only have Friday afternoons. Cutting scope is the main fix.

Risk or decision	Options	My recommendation
Too much to build	Ship the ten MVP sections only	Cut Plus, Momentum and sharing from v1
Data entry for degree plans	Hand-enter one or two programs	Start with your own program
Scraping public pages	Ask permission, or skip	Skip until you have a written yes
App name	Koalified or Pouch	Pick one and check the App Store for clashes
Plus price	$X.XX per semester	Decide after you know the AI coach cost per user

Risk or decision	Options	My recommendation
Streak rules (later)	Strict, rest day, or grace window	Rest day, to avoid punishing exam weeks
Android	Later or never	Later, only if students ask
Accounts	Not in v1	Add only when sync or buddies need them

- [ ] Read UQ's policies and email Student Services or ITS about public pages
- [ ] Confirm the IP position on your project
- [ ] Confirm the koala artwork rights in writing
- [ ] Pick the app name
- [ ] Enter real data for your own program and semester dates
- [ ] Set up the Apple Developer account
- [ ] Recruit 20 TestFlight testers
- [ ] Write the privacy policy and support page